FM Stereo Encoder Build and Test Guide

Build the board in 12 stages. Solder one block, test it, and only then move to the next one. If a test fails, stop and fix it before you add more parts. This guide matches the Encoder_gen board.

Before you start

- Tools you need are a soldering iron, a multimeter, a scope with two channels, a function generator and a bench supply with a current limit.
- Set the supply to plus 9V and minus 9V with the current limit at 100 mA. Raise it only if a stage says so.
- Solder every IC into a socket. Leave the sockets empty until a stage tells you to insert the chip.
- Test points (TP) are bare pads. There is nothing to solder on them. Touch the probe tip to the pad and clip the probe ground to any TP_GND pad.
- Use a 1 kHz sine at 1 Vpp with no DC offset as your audio test signal, unless a stage says otherwise.
- Insert ICs with the power off. Pin 1 goes at the notch on the socket.
- Write every reading in the log at the end. If something fails later, the log shows when it started.

Where the connectors are

- J1 is the minus 9V input. Pin 1 is minus 9V and pin 2 is GND.
- J2 is the plus 9V input. Pin 1 is GND and pin 2 is plus 9V.
- J3 is the audio input. Pin 1 is left, pin 2 is right, pin 3 is not used and pin 4 is GND.
- J4 is the output to your FM module. Pin 1 is the MPX output and pin 4 is GND.
- J5 is the BNC for your 19 kHz sine.
- JP1 is the 3 pin carrier jumper. A shunt on pins 1 and 2 is the normal carrier. A shunt on pins 2 and 3 flips the carrier.

The 12 stages

1. Power supply
2. Input protection and AC coupling
3. Left and right buffers with filters (U1 and U3)

4. Mixing buffer (U4)
5. Inverting stage (U5)
6. Summing amplifier (U6)
7. PLL 38 kHz carrier generator and BNC (U11, U12, U13)
8. Balanced modulator (U9)
9. Difference to single ended amplifier (U10A)
10. MPX summer with pilot (U8)
11. Output filters (U10B and U2) and J4
12. Full system test and pilot level

Stage 1 Power supply

Solder

- D1 and D2 are 1N5819 diodes. The striped end (K) goes to the square pad.
- C3 and C4 are 22u electrolytic capacitors. The long lead goes to the square pad marked plus on the silk. Wrong polarity can make them burst.
- J1 and J2 are the screw terminals. The wire entry faces the board edge.

Test

1. Before you connect the board, check your supply leads with the multimeter. J1 pin 1 needs minus 9V against GND. J2 pin 2 needs plus 9V against GND.
2. Connect the supply with the current limit at 100 mA. The current should read close to 0 mA.
3. Measure TP_+9V1 against GND. You should read about plus 8.6 to plus 9.0V. The Schottky diode drops 0.2 to 0.4V, so the rail sits slightly below your supply setting.
4. Measure TP_-9V1 against GND. You should read about minus 8.6 to minus 9.0V.
5. With the power off, check continuity from J1 pin 2 to every GND test point, J3 pin 4, J4 pin 4 and the BNC shell once it is fitted. All of them must beep.

If it fails

- A rail at 0V means a diode is backwards or a terminal joint is cold.
- A rail that is far too low and the current limit lights means a capacitor is backwards.

Stage 2 Input protection and AC coupling

Solder

- R2 and R3 are 10k.
- R4 and R6 are 100k.
- C11 and C12 are 100p ceramic.
- D3, D4, D5 and D6 are 1N4148. The striped end (K) goes to the square pad.
- C1 and C2 are 1u film capacitors.
- J3 is the 4 pin female header.

Test

1. Power on with no ICs fitted anywhere.
2. Measure TP_L1 and TP_R1 against GND. Both should read 0V within 5 mV.
3. Connect the generator to J3 pin 1 and pin 4 (1 kHz, 1 Vpp). TP_L1 should show a clean sine of about 0.9 Vpp.
4. Move the generator to J3 pin 2 and pin 4. TP_R1 should show about 0.9 Vpp and TP_L1 should show nothing.
5. Raise the generator to 3 Vpp. The sine at TP_L1 should stay clean and grow to about 2.7 Vpp.

If it fails

- No signal at all means C1 or C2 is not soldered, or R2 is open.
- An amplitude near 1 Vpp with a distorted shape means a diode is backwards.

Stage 3 Left and right buffers with filters (U1 and U3)

Solder

- U1 and U3 get DIP 8 sockets.
- C7, C8, C9, C13 and C14 are 100n.
- C23, C24, C25 and C26 are 1n.
- R9, R10, R11 and R12 are 10k.

Test

1. Power off. Insert a TL072 in U1 and in U3.
2. Power on. On each chip measure pin 8 against GND for plus 9V and pin 4 for minus 9V.

3. Left channel. Feed 1 kHz and 1 Vpp to J3 pin 1 and pin 4. TP13 (U1A_OUT) and TP15 (U1B_OUT) should each show about 0.9 Vpp and be in phase with TP_L1.
4. Right channel. Feed J3 pin 2 and pin 4. TP14 (U3A_OUT) and TP16 (U3B_OUT) should each show about 0.9 Vpp.
5. DC at all four test points should read within 20 mV of 0V.
6. Raise the generator to 10 kHz. TP15 and TP16 should fall to about 0.65 Vpp.

If it fails

- A flat output stuck near plus or minus 8V means that op amp has no feedback. Check the pin 1 to pin 2 link or pin 6 to pin 7 link on the board.
- A hot chip means it sits backwards in the socket. Power off at once.

Stage 4 Mixing buffer (U4)

Solder

- U4 gets a DIP 8 socket.
- C15, C43 and C44 are 100n.
- C31 and C33 are 330p.
- C32 and C34 are 2.2n.
- R13, R14, R15 and R17 are 12k.
- R19, R20, R21, R22 and R23 are 100k. They sit in this block but feed the next stages.

Test

1. Power off. Insert a TL072 in U4. Power on.
2. Feed J3 pin 1 and pin 4 (1 kHz, 1 Vpp). TP_L2 should show about 0.9 Vpp. TP_L2 is the left output of U4. TP17 (U4A_OUT, the right output) should show nothing.
3. Move the generator to J3 pin 2 and pin 4. TP17 should show about 0.9 Vpp and TP_L2 should show nothing.
4. Sweep the generator on the left input and watch TP_L2. The calculated response of U1 and U4 together is about 0.6 dB down at 10 kHz, 3 dB down at 15 kHz and 8 dB down at 19 kHz. Allow 1.5 dB for part tolerance.
5. Check DC at TP_L2 and TP17. Both should read within 20 mV of 0V.

If it fails

- A large peak near 15 kHz means a filter capacitor value is wrong. Recheck C31 to C34.
- No output on one side means the 12k resistor in front of that op amp is missing.

Stage 5 Inverting stage (U5)

Solder

- U5 gets a DIP 8 socket.
- C10 and C16 are 100n.
- R24, R26 and R28 are 100k.
- R32 is 51 ohms.

Test

1. Power off. Insert a TL072 in U5. Power on.
2. Left only. Feed J3 pin 1 and pin 4 (1 kHz, 1 Vpp). TP18 (U5A_OUT) should show about 0.9 Vpp and be upside down against TP_L2. TP19 (U5B_OUT) should show about 0.9 Vpp and be in phase with TP_L2.
3. Same signal on both channels. Join J3 pin 1 and pin 2 together and feed them against pin 4. TP18 and TP19 should both drop to nearly nothing, under 50 mVpp. This proves the left minus right subtraction works.
4. DC at TP18 and TP19 should read within 20 mV of 0V.

If it fails

- If TP19 stays at 0.9 Vpp when you join both inputs, R20 or R21 is the wrong value or missing.

Stage 6 Summing amplifier (U6)

Solder

- U6 gets a DIP 8 socket.
- C21 and C22 are 100n.
- R25, R27 and R30 are 100k.
- R45 and R48 are 100k.
- R47 is 360k. EKT has no 360k, so use a 330k and a 30k in series. Put the 330k in one R47 hole and twist one 30k lead onto the free lead of the 330k in the air above the board. Solder that joint and slide on a piece of sleeving. Put the other 30k lead in the second R47 hole.

Test

1. Power off. Insert a TL072 in U6. Power on.

2. Left only. Feed J3 pin 1 and pin 4 (1 kHz, 1 Vpp). TP20 (U6A_OUT) should show about 0.9 Vpp upside down against TP_L2. TP21 (U6B_OUT) should show about 0.9 Vpp in phase.
3. Same signal on both channels. Join J3 pin 1 and pin 2 and feed them against pin 4. TP21 should now show about 1.8 Vpp. This proves the left plus right addition works.
4. DC at TP20 and TP21 should read within 20 mV of 0V.
5. With the power off, measure R47 across its two pads. It should read about 360k.

If it fails

- A TP21 reading of 0.9 Vpp with both inputs joined means one channel into U6A is missing. Check R22 and R23.

Stage 7 PLL 38 kHz carrier generator and BNC

This block turns your 19 kHz sine into a locked 38 kHz carrier. Test it alone before you build the modulator.

Solder

- U11 gets a DIP 16 socket. U12 and U13 get DIP 14 sockets.
- R51 is 12k. R52 is 10k. R53 is 4.7k. R54 is 8.2k. R55 is 1.5k.
- C47 is 1n. C48, C50, C51 and C52 are 100n.
- C45, C46 and C49 are 1u film capacitors.
- J5 is the BNC. JP1 is the 3 pin male header.

Test with the chips out

1. Power on with U11, U12 and U13 empty.
2. Measure the sockets. U11 pin 16 and pin 14 on U12 and U13 should read plus 9V. U11 pins 5 and 8, and pin 7 on U12 and U13, should read 0V.
3. Power off.

Test with the chips in

1. Insert the CD4046 in U11, the CD4013 in U12 and the CD4069 in U13. Power on.
2. The supply current should rise by less than 15 mA. If it is higher, power off and look for a backwards chip.
3. Fit a shunt on JP1 pins 1 and 2. Leave J5 empty.
4. Probe TP33 (38K_VCO). You should see a square wave from 0V up to about 8.7V. It may drift. Write down its frequency. A reading between 20 kHz and 100 kHz is healthy.

5. Probe TP34 (19K_DIV). It should be a square wave at exactly half the TP33 frequency.
6. Probe TP12 (38K_IN). You should see a square wave of about 1.3 Vpp centred on 0V.
7. Connect your generator to J5 and set 19 kHz, 1 Vpp, no offset.
8. Probe TP11 (19K_IN). It should show the same 19 kHz sine.
9. Probe TP33 and TP34 together. TP33 must settle at exactly 38 kHz and TP34 at exactly 19 kHz. Trigger the scope on TP34. When locked, TP33 stands still on the screen.
10. Move the generator 500 Hz up and down. TP33 should stay at exactly twice the generator setting.
11. Pull the BNC cable out. TP33 should leave 38 kHz. Plug it back in and it should return. This proves the PLL does the work.
12. Move the shunt to JP1 pins 2 and 3. TP12 should flip upside down against TP33. Move it back to pins 1 and 2.

If it fails

- No square wave at TP33 means R51 or C47 is wrong, or U11 pin 16 has no supply.
- A wave at TP33 that never locks means the VCO range does not reach 38 kHz. If your free running reading in step 4 was below 30 kHz, change R51 to 10k. If it was above 80 kHz, change R51 to 15k.
- TP34 not at half of TP33 means U12 is not dividing. Check U12 pin 14 and the clock link to TP33.
- A weak lock that drops out means the 19 kHz level is too low. Raise it to 2 Vpp.

Stage 8 Balanced modulator (U9)

Solder

- U9 gets a DIP 14 socket.
- R29, R35 and R37 are 1k.
- R31 is 51 ohms.
- R33 is 6.8k.
- R34 and R36 are 3.9k.
- R38 is 10k.
- R41 is 7.5k.
- R42 is 2.2k.
- C37, C38, C39, C40, C41 and C42 are 100n.

Test the DC levels

1. Power off. Insert the MC1496 in U9. Power on.
2. Measure these pins against GND.
3. Pin 14 should be about minus 8.7V.
4. Pin 5 should be about minus 8.0V.
5. Pins 8 and 10 should be about plus 2.0V.
6. Pins 6 and 12 should each be about plus 6.4V, and equal to each other within 0.2V.
7. Pins 1 and 4 should be close to 0V.

Test the signal

1. Keep the PLL locked to 19 kHz at J5. Probe TP22 (MOD_OUT_P) and TP23 (MOD_OUT_N). With no audio they show only a small 38 kHz leakage. Write the amplitude down.
2. Feed left only (J3 pin 1 and pin 4, 1 kHz, 1 Vpp). TP22 and TP23 should now show a 38 kHz carrier whose height rises and falls with the 1 kHz audio. It should shrink to nearly zero at each audio zero crossing.
3. TP22 and TP23 must be upside down against each other.
4. Join J3 pin 1 and pin 2 (the same signal on both channels). The 38 kHz should almost vanish. It should fall at least 20 dB below the left only reading. This proves the balance is good.

If it fails

- A pin 5 reading far from minus 8V means R33 is wrong.
- Pins 6 and 12 that differ by more than 0.5V mean R34 or R36 is the wrong value or a joint on pin 6 or 12 is cold.
- A strong carrier with no audio means the MC1496 is hot or the two pin 8 and pin 10 levels differ. Check R35, R37, R41 and R42.

Stage 9 Difference to single ended amplifier (U10A)

Solder

- U10 gets a DIP 8 socket.
- C29 and C30 are 100n.
- R39, R40, R43 and R44 are 15k.
- R46 is 270k.

Test

1. Power off. Insert an NE5532 in U10. Power on.
2. Keep the PLL locked and feed left only (1 kHz, 1 Vpp).
3. Probe TP24 (U10A_OUT). You should see the same 38 kHz carrier shape that you saw on TP22 and TP23, now as one signal. Its height should be about the difference between the two (close to double one side).
4. Join J3 pin 1 and pin 2. TP24 should almost vanish, like in stage 8.
5. DC at TP24 should read within 50 mV of 0V.

If it fails

- No signal at TP24 means R39 to R44 has a wrong value or an open joint.
- A heavy DC offset at TP24 means the four 15k resistors are not matched. Check each with the meter.

Stage 10 MPX summer with pilot (U8)

Solder

- U8 gets a DIP 8 socket.
- C28 is 100n.
- R49 and R50 are 100k.

Test

1. Power off. Insert a TL072 in U8. Power on.
2. Audio only. Take the BNC cable out. Join J3 pin 1 and pin 2 and feed 1 kHz at 1 Vpp. TP26 (U8A_OUT) should show about 1.8 Vpp and be in phase with TP21. TP27 (U8B_OUT) should show the same sine upside down.
3. Pilot only. Remove the audio. Feed 19 kHz at 1 Vpp to J5. TP26 should show a 19 kHz sine of about 0.28 Vpp. The gain is 100k divided by 360k. Allow 10 percent.
4. DC at TP26 and TP27 should read within 30 mV of 0V.

If it fails

- A pilot much smaller than 0.28 Vpp means R47 is too large. Measure it again.
- A pilot much larger than 0.28 Vpp means R47 is too small, or C46 is shorted.

Stage 11 Output filters (U10B and U2) and J4

Solder

- U2 gets a DIP 8 socket.
- C18, C19 and C27 are 100n.
- R1 and R5 are 2.4k. EKT has no 2.4k, so use 2.2k if you accept the small change in filter frequency.
- R7 and R8 are 1.8k.
- R16 and R18 are 3.3k.
- C5 is 1n.
- C6 is 1.2n. EKT has no 1.2n, so solder a 1n and a 220p in parallel on the C6 pads.
- C17 is 2.2n. C20 is 1n. C35 is 3.3n. C36 is 220p.
- J4 is the 4 pin female header.

Test

1. Power off. Insert an NE5532 in U2. Power on.
2. Join J3 pin 1 and pin 2 and feed 1 kHz at 1 Vpp. Probe TP25 (U10B_OUT), TP28 (U2A_OUT) and TP29 (OUT). All three should show about 1.8 Vpp, within 5 percent of TP26.
3. Sweep the generator on the audio input. The three filters pass the full MPX band. The calculated response is within 2 dB up to 53 kHz and at least 20 dB down at 100 kHz.
4. DC at TP29 should read within 20 mV of 0V.
5. The audio filters in front of the matrix cut at 15 kHz, so you can only check the response above 15 kHz by feeding the pilot. Feed 19 kHz at 1 Vpp to J5 with no audio. TP29 should show about 0.27 Vpp.

If it fails

- A very large peak above 40 kHz means C35 or C36 is the wrong value.
- A TP29 level well below TP26 means R16 or R18 is wrong.

Stage 12 Full system test and pilot level

Set the pilot level

1. Fit every part and every chip. Power on.

2. Feed your loudest audio, for example J3 pin 1 and pin 2 joined at 1 Vpp. Take the BNC cable out and read the peak to peak at TP29. Call that value Vpp.
3. The pilot should be about 9 percent of the full signal. Set the generator at J5 to 0.32 times Vpp. For example, 1.8 Vpp at TP29 means about 0.58 Vpp at J5.
4. Plug the BNC back in. Check that the PLL still locks.

Check the stereo phase

1. Connect your FM transmitter module to J4 pin 1 and pin 4. Check the maximum audio level of the module first. Never exceed it.
2. Tune an FM receiver to the transmitter.
3. Feed a 1 kHz tone to the left input only (J3 pin 1). You must hear it in the left ear only.
4. If you hear it in the right ear, power off, move the JP1 shunt to the other position and test again.
5. Repeat with the right input only.

Final checks

- The receiver should show the stereo light.
- Feed the same tone to both inputs. The sound should come out of both sides with no hiss or whine.
- Let the board run for 10 minutes. No part should be warm.

Readings log

Fill the Measured column as you build. Add rows for anything else you check.

Stage	Test point	Expected	Measured
1	TP_+9V1	plus 8.6 to 9.0V	
1	TP_-9V1	minus 8.6 to 9.0V	
2	TP_L1	0.9 Vpp	
2	TP_R1	0.9 Vpp	
3	TP15	0.9 Vpp	
3	TP16	0.9 Vpp	
4	TP_L2	0.9 Vpp	
4	TP17	0.9 Vpp	

Stage	Test point	Expected	Measured
5	TP19 with both inputs joined	under 50 mVpp	
6	TP21 with both inputs joined	1.8 Vpp	
7	TP33 free running	20 to 100 kHz	
7	TP33 locked	38 kHz exactly	
7	TP34 locked	19 kHz exactly	
8	U9 pin 5	about minus 8.0V	
8	U9 pins 6 and 12	about plus 6.4V	
9	TP24	DSB carrier, nearly zero when both inputs are joined	
10	TP26 pilot only	0.28 Vpp	
11	TP29 audio	1.8 Vpp	
12	Pilot at J5	0.32 times the audio Vpp	

Parts that EKT does not stock

- 360k for R47. Use a 330k and a 30k in series, as stage 6 shows.
- 2.4k for R1 and R5. Use 2.2k, or ask EKT for 2.4k.
- 1.2n for C6. Use a 1n and a 220p in parallel.
- 1u film capacitors may be larger than the board footprint. Measure the part before you solder. If it does not fit, bend the leads or ask EKT for a smaller 1u part.
- The 22u electrolytics (C3 and C4) are 5 mm wide and the footprint is for 8 mm. Bend the leads to reach the holes.